ALL EYES ON U.S.: REGULATING THE USE & DEVELOPMENT OF FACIAL RECOGNITION TECHNOLOGY

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ABSTRACT

Facial recognition technology is developing faster than laws are being passed to properly regulate it. Because of its chilling effect on Americans' First Amendment rights, risk to international data transfers, demonstrated biased outcomes, and threat to our most sensitive data, the U.S. must take steps toward effectively controlling its use. This Article is aimed at first, addressing the history and social context surrounding facial recognition technology development and its use, and second, analyzing and identifying best practices that legislators, scholars, and corporate policymakers alike can take into account in considering robust protective measures to shield our most precious class of data.

I. INTRODUCTION

2020 was a year defined by tumult on a global scale. Much of the adversity the world faced was caused by what many perceived as a series of firsts, but in reality, all of the worst parts of the year stemmed from problems that have existed for centuries. Three hardships in particular stood out: (1) the death of George Floyd and the global protests that erupted as a result,[1] (2) the role of facial recognition technology and the U.S. response to it during a racially charged time, and (3) a global pandemic raging in the background. While some may argue that it was the combination of these three factors that created the novelty of the year 2020, it was not so long ago that the U.S. was faced with the convergence of parallel factors.

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[1] Zamira Rahim & Rob Picheta, *Thousands Around the World Protest George Floyd's Death in Global Display of Solidarity*, CNN (June 1, 2020, 6:31 PM), https://www.cnn.com/2020/06/01/world/george-floyd-global-protests-intl/index.html.

In fact, in the 1960's, the U.S. encountered the civil rights movement, the advancement of facial recognition technology,[2] and a deadly global flu pandemic that originated in China and claimed more lives than the coronavirus.[3] The critical distinction, however, is in the heightened collective social conscience that has drawn clearer connections between these factors, especially facial recognition technology and racial bias. Further, as individuals have become more concerned with their human rights, including their right to privacy, there have been commensurate developments in the law to reflect this. In many ways, these developments can serve as a framework for drafting a much-needed biometric data privacy law to ensure that our most precious data is proportionally protected.

Biometric data is critical because once compromised, there is no effective remediation that can undo the potential damage caused. In 2020, the convergence of racial tension in the U.S., following George Floyd's death, and increasing self-awareness of data subject rights, following the enactment of the EU's General Data Protection Regulation (GDPR), brought facial recognition's premature uses to the forefront. While Americans exercised their First Amendment right to protest, joined shortly thereafter by the rest of the world, some companies joined in the fight by changing corporate policies to turn away from biased technologies.

The U.S. is at an inflection point from a regulatory perspective, where it is facing expectations set by domestic social standards and global pressures alike. The current fragmented regulatory environment is simply ineffective and dysfunctional. For the sake of streamlined, feasible compliance, and ensuring continued international economic prosperity, the U.S. must re-examine and reinforce its federal statutory infrastructure to meet its current challenges in the privacy space; especially when it comes to sensitive

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[2] Shaun Raviv, *The Secret History of Facial Recognition*, WIRED (Jan. 21, 2020, 6:00 AM), https://www.wired.com/ story/secret-history-facial-recognition/.

[3] Bojan Pancevski, *Forgotten Pandemic Offers Contrast to Today's Coronavirus Lockdowns*, WALL ST. J. (Apr. 24, 2020, 5:30 AM), https://www.wsj.com/articles/forgotten-pandemic-offers-contrast-to-todays-coronavirus-lockdowns-11587720625.

biometric data. This Article will first explain the historical background leading up to the present challenges in data privacy and then analyze and propose best practices for legislators, scholars, and corporate policymakers alike to consider.

II. BACKGROUND

A. The Inextricable Relationship Between Racial Tension in America & Facial Recognition Technology

a. The First Attempts at Facial Recognition Technology

Facial recognition has made its way into the global mainstream in recent years due to its rapid advancements. Some of its most common consumer-facing applications are used via security features and digital image filters. From the security standpoint, the advancement of the technology can be illustrated through the example of Apple, Inc.'s response to COVID-19 as it began testing software to permit iPhone users to access their devices through facial recognition, even if the individual is wearing a face mask.[4] However, what is less frequently discussed are the origins of the technology, which can be traced back over half a century ago and only a few years following the coining of the term "Artificial Intelligence."[5]

In the 1960's, a man named Woodrow "Woody" Wilson Bledsoe ("Bledsoe") first identified facial recognition as a viable biometric.[6] In partnership with his colleague, Iben Browning ("Browning"), Bledsoe co-created the *n*-tuple recognition method; a type of weightless neural network, which is a practical pattern recognition method based on distributed computation and amenable to

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[4] Lauren Goode, *The iPhone's Face ID Will Soon Work with a Mask—if You Have an Apple Watch*, WIRED (Feb. 2, 2021, 7:00 AM), https://www.wired.com/story/iphone-face-id-mask-ios-beta/.

[5] Gil Press, *A Very Short History of Artificial Intelligence (AI)*, FORBES (Dec. 30, 2016, 9:09 AM), https://www. forbes.com/sites/gilpress/2016/12/30/a-very-short-history-of-artificial-intelligence-ai/?sh=727f48d56fba.

[6] *See* Raviv, *supra* note 2.

description in terms of neural network metaphors.[7] To this day, the *n*-tuple method continues to be recognized for its speed and simplicity when analyzing certain large data sets.[8] The analysis Bledsoe and Browning conducted is simpler to understand with an illustrative example; one of their first demonstrations leveraged alphabetical character recognition:

> They started by projecting a printed character—the letter *Q*, say—onto a rectangular grid of cells, resembling a sheet of graph paper. Then each cell was assigned a binary number according to whether it contained part of the character: Empty got a 0, populated got a 1. Then the cells were randomly grouped into ordered pairs, like sets of coordinates. (The groupings could, in theory, include any number of cells, hence the name *n*-tuple.) With a few further mathematical manipulations, the computer was able to assign the character's grid a unique score. When the computer encountered a new character, it simply compared that character's grid with others in its database until it found the closest match.[9]

These advancements allowed for a wider application of pattern recognition, beyond just letters and characters,[10] effectively serving as the foundation of facial recognition technology.

Two particularly noteworthy aspects of Bledsoe and Browning's discoveries were: first, they asked the question of "how can we make a machine do something like what people do?"[11] Secondly, however,

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[7] Richard Rohwer & Michal Morciniec, *A Theoretical and Experimental Account of n-tuple Classifier*, 8:3 NEURAL COMPUTATION 629 (2008).
[8] *Id.*
[9] *See* Raviv, *supra* note 2.
[10] *Id.*
[11] Anne O. Boyer & Robert S. Boyer, *A Biographical Sketch of W.W. Bledsoe*, *in* AUTOMATED REASONING: ESSAYS IN HONOR OF WOODY BLEDSOE 11 (Robert S. Boyer ed., 1991).

these critical first steps into the examination of the math and science behind pattern recognition were framed by a backdrop of civil rights activism. When harkening back to the 1960's, it is not facial recognition technology that defined the zeitgeist. Instead, it was a time of protest with unprecedented, iconic, history-shaping moments like those brought about by the Freedom Riders,[12] the March on Washington,[13] the passage of the Civil Rights Act of 1964,[14] the signing of the Voting Rights Act of 1965,[15] and the assassination of major civil rights figures of the era, including Malcolm X, Dr. Martin Luther King, Jr., and Robert F. Kennedy, among many others.[16]

It is also important to note the intended application of facial recognition technology. Bledsoe, Browning, and another colleague formed Panoramic Research, Inc. ("Panoramic") and its primary source of funding came from the U.S. government via the Central Intelligence Agency ("CIA"); what is more, the CIA helped keep Panoramic financially viable when the team proved unsuccessful at raising funding through alternate channels.[17] At the time when Panoramic started working with them, the CIA was engaging in questionable, unethical, illegal experiments—among one of the most infamous of these experiments was Project MK-Ultra, the U.S. government's attempt at "mind control" through the psychological

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[12] RAYMOND ARSENAULT, FREEDOM RIDERS: 1961 AND THE STRUGGLE FOR RACIAL JUSTICE 3 (2007) (ebook) (discussing the "Freedom Riders," a biracial group of individuals who faced police violence head-on while protesting segregation in America).

[13] *See* Lucy G. BARBER, MARCHING ON WASHINGTON: THE FORGOING OF AN AMERICAN POLITICAL TRADITION 534-35 (2004) (discussing "the March on Washington," a gathering of over 200,000 people of all races congregating to hear Dr. Martin Luther King, Jr.'s "I Have a Dream" speech).

[14] 42 U.S.C.A. § 2000e (West 1964) (a law guaranteeing equal employment, limiting the use of voter literacy tests, and ensuring that public facilities were integrated).

[15] Voting Rights Act of 1965, Pub. L. No. 89-110, 79 Stat. 437, 439 (banning voter literacy tests and carving a path towards poll taxes being declared unconstitutional via *Harper v. Virginia*).

[16] Harper v. Va. State Bd. of Elections, 383 U.S. 663, 670 (1966).

[17] *See* Raviv*, supra* note 2, at 9.

torture of unwilling participants.[18] But it was in 1963 that Bledsoe would successfully pitch a minimal viable product ("MVP"), a facial recognition machine capable of recognizing ten faces in the hopes of eventually being able to scale the MVP's capability up to analyze and recognize thousands of faces.[19] The facial recognition machine proved mostly unsuccessful. However, between that attempt and another government-funded assignment that Bledsoe would take on later in 1967 aimed at assisting the police with identifying faces,[20] Bledsoe made objectively forward progress towards developing a functional prototype, although at the time technological limitations represented a significant constraint. These limitations surfaced in a number of ways, for instance, the three-dimensional, non-static features of a face and its ability to distort substantially through movements as simple as a smile.[21]

While Bledsoe deserves to be lauded for his scientific accomplishments, it should also be acknowledged that the problems that are systemic across recent attempts at developing facial recognition technology today started with Bledsoe as well. Namely, in addition to colluding with—and developing the technology for the use of—corrupt government ventures, journalist Shaun Raviv ("Raviv") summarized the poor ethical standard Bledsoe helped establish, which continues to plague the industry in the modern era. Effectively, Raviv stated "the sample sets skewed almost entirely toward white men; the seemingly blithe trust in government authority; the temptation to use facial recognition to discriminate between races—continue to dog the technology today."[22]

b. Recognizing a Pattern of Enduring Bias

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[18] Project MKUltra, the CIA's Program of Rsch. in Behav. Modification: Joint Hearings before the Select Comm. on Intel. and the Subcomm. on Health & Sci. Rsch. of the Comm. on Hum. Res., 95th Cong. (1977).
[19] *See* Raviv, *supra* note 2, at 11.
[20] *Id.* at 17.
[21] *Id.* at 16.
[22] *Id.* at 20.

2020 was a year when facial recognition technology made its way to the forefront of the mainstream while, once again, tumultuous race relations raged in the background. Although the technology has steadily improved over the last few years, and while racial tensions in America have existed since the nation's inception, June 2020 is when these two seemingly unrelated elements collided in a very critical way. In the prior month, George Floyd died after being handcuffed and pinned to the ground by a Minneapolis police officer's knee, following a call by a store clerk claiming that Mr. Floyd, a black man, had paid for cigarettes with a counterfeit twenty-dollar bill.[23] After Mr. Floyd's death, the whole world erupted in protest.[24] Outside of Minneapolis, all across the globe, average individuals, scholars, and experts seemed to be applying intense scrutiny to institutional practices broadly across industries to uncover areas potentially in need of reform.[25] One such example was facial recognition technology and the biases that underpin it.

In a study published by the National Institute of Standards and Technology ("NIST"), the institute concluded that despite the high accuracy rate reported by modern facial recognition solutions, there was still substantial bias in the outcomes.[26] Among its findings, NIST disclosed that for "law enforcement images, it is empirically observed that men are more easily recognized than women, that heavier individuals are more easily recognized than lighter subjects, and that Asian subjects are more easily recognized than White. Younger persons are more difficult to recognize than their elders for some recognition algorithms, but the opposite is true for others."[27] In a more recent 2019 study, NIST concluded that algorithms falsely identified African-American and Asian faces 10 to 100 times more

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[23] *How George Floyd Died, and What Happened Next*, N.Y. TIMES (May 25, 2021), https://www.nytimes.com/ article/george-floyd.html.
[24] *See* Rahim & Picheta, *supra* note 1.
[25] *Id.*
[26] Patrick J. Grother et al., Nat'l Inst. of Standards & Tech., Report on the Evaluation of 2D Still-Image Face Recognition Algorithms 55 (2011), https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir7709.pdf.
[27] *Id.*

than Caucasian faces.[28] In yet another study, it was found that facial recognition accuracy is poorest for subjects who are female, Black, and 18-30 years old.[29] These issues around inaccuracy may at first appear marginal and perhaps even relatively inconsequential, but that is only the case when examined in a vacuum. To illustrate the significance of this problem, one must first recognize that incarceration rates in the U.S. have historically and consistently skewed heavily towards African Americans and other minorities.[30] Naturally, therefore, "[b]lack people are overrepresented in mugshot data, which face recognition uses to make predictions."[31] Thus, the use of facial recognition technology heightens the risk of exposure for future bias and contributes to increasing the disproportionate incarceration rate.

As the George Floyd protests continued throughout June 2020 on the nation's main stage, in the background a separate clash was playing out via newly escalated questions around the threat of facial recognition to First Amendment rights. Specifically, Congressional Democrats conducted probes to determine the extent of U.S. federal agencies' reliance on facial recognition technology to wrongfully track protesters.[32] It had already been well-documented that the

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[28] Natasha Singer & Cade Metz, *Many Facial-Recognition Systems Are Biased, Says U.S. Study*, N.Y. TIMES (Dec. 19, 2020), https://www.nytimes.com/2019/12/19/technology/facial-recognition-bias.html.

[29] Brendan F. Klare et al., *Face Recognition Performance: Role of Demographic Information*, 7 IEEE TRANSACTIONS ON INFO. FORENSICS AND SEC. 1789, 1800 (2012).

[30] Brett E. Garland et al., Racial Disproportionality in the American Prison Population: Using the Blumstein Method to Address the Critical Race and Justice Issue of the 21st Century, 5 JUST. POL'Y J. 1, 4 (2008).

[31] Alex Najibi, *Racial Discrimination in Face Recognition Technology*, SITN (Oct. 24, 2020), https://sitn.hms.harvard.edu/flash/2020/racial-discrimination-in-face-recognition-technology/.

[32] Irina Ivanova, *Why Face-Recognition Technology Has a Bias Problem*, CBS (June 12, 2020, 7:57 AM), https://www.cbsnews.com/news/facial-recognition-systems-racism-protests-police-bias/; *See* Anna G. Eshoo et al., *Congressional Letter to Federal Bureau of Investigation (FBI), the National Guard Bureau, the Drug Enforcement Administration (DEA), and Customs and Border Protection (CBP)*, U.S. HOUSE OF REPRESENTATIVES (June 9, 2020), https://eshoo.house.gov/sites/eshoo.house.gov/files/Eshoo-

police had been engaging in these types of activities since at least 2015 when individuals flooded the streets of Baltimore to protest the death of Freddie Gray, a young Black man who died in police custody from spinal injuries, for which no one was held responsible.[33] Companies like Clearview AI illustrate the unfettered means of contributing to this type of tracking by scraping publicly available images at a massive scale to feed into a database to sharpen its algorithms[34] – a technique enabled by current technologies that Bledsoe could only ever have dreamed of in the 1960s. Companies like Clearview AI operate in a federally unregulated environment and, therefore, exercise a wide latitude through their means of data collection. As stated by the Vermont Attorney General in a recently filed lawsuit against Clearview AI in Vermont, "I am disturbed by [Clearview AI's] practice[s], particularly the practice of collecting and selling children's facial recognition data . . . . This practice is unscrupulous, unethical, and contrary to public policy."[35] Clearview AI alleges that the primary purpose of its offerings is for law enforcement usage, and this includes a partnership with the Minneapolis Police Department – the same police department that was held accountable for the death of George Floyd, following recent civil[36] and criminal[37] rulings.

Rush%20Ltr%20to%20FBI%2C%20NG%2C%20CBP%2C%20DEA%20on%20government%20surveillance%20of%20protesters%20-%206.9.20.pdf.

[33] Malkia Devich-Cyril, *Defund Facial Recognition*, THE ATLANTIC (July 5, 2020), https://www.theatlantic.com/technology/archive/2020/07/defund-facial-recognition/613771/.

[34] Aaron Holmes, Vermont's Attorney General Is Suing A Controversial Facial Recognition App That Uses Images Scraped from Social Media to Create A Database for Police and Wealthy Clients: 'I Am Disturbed,' BUS. INSIDER (Mar. 11, 2020, 11:02 AM), https://www.businessinsider.com/clearview-ai-vermont-attorney-general-lawsuit-facial-recognition-2020-3.

[35] *Id.*

[36] Nicholas Bogel-Burroughs & John Eligon, *George Floyd's Family Settles Suit Against Minneapolis for $27 Million*, N.Y. TIMES (Mar. 30, 2021), https://www.nytimes.com/2021/03/12/us/george-floyd-minneapolis-settlement.html.

[37] Eric Levenson & Aaron Cooper, *Derek Chauvin Found Guilty of All Three Charges for Killing George Floyd*, CNN (Apr. 21, 2021, 12:13 PM),

The current critical consensus on facial recognition technology can be best summarized by the American Civil Liberties Union's ("ACLU") Kade Crockford, who testified before the Boston City Council stating: "Face surveillance is dangerous when it works and when it doesn't."[38] In other words, inaccuracy can lead to poor outcomes including biased analyses and resultingly disproportionate incarcerations and practices that generally disfavor minority groups. However, even if the technology was to function with complete accuracy, in the currently unregulated U.S. environment, "it could spell the end of privacy as Americans know it."[39] This is why many progressive companies within the technology sector have already taken proactive action to slow or entirely halt their activities in the space. For example, in June 2020, as part of a series of broad-sweeping policy changes to create a more inclusive culture at IBM, a global fortune 100 tech company, CEO Arvind Krishna penned a letter to Congress and issued press releases wherein he explained that the company would be halting the research, development, and distribution of this kind of facial recognition technology.[40] In his statement, Mr. Krishna wrote:

> IBM firmly opposes and will not condone uses of any [facial recognition] technology, including facial recognition technology offered by other vendors, for mass surveillance, racial profiling, violations of basic human rights and freedoms, or any purpose which is not consistent with our values and Principles of Trust and Transparency . . . . We believe now is the time to begin a national dialogue on whether and how facial recognition technology

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https://www.cnn.com/2021/04/20/us/derek-chauvin-trial-george-floyd-deliberations/index.html.

[38] *See* Ivanova, *supra* note 32.

[39] *Id.*

[40] *IBM CEO's Letter to Congress on Racial Justice Reform*, IBM THINKPOLICY BLOG (June 8, 2020), https://www.ibm.com/blogs/policy/facial-recognition-sunset-racial-justice-reforms/.

> should be employed by domestic law enforcement agencies.[41]

Shortly thereafter, other large technology companies came forward to express similar sentiments in an attempt to self-regulate in the absence of consistent, robust federal guidance. For example, in the same week as IBM's announcement, Amazon also set a self-imposed one-year moratorium on police use of its facial recognition software, known as Rekognition.[42] In its statement, Amazon "advocated that governments should put in place stronger regulations to govern the ethical use of facial recognition technology."[43] Prior to this statement, an MIT study had found that "Amazon's facial analysis software distinguishes gender among certain ethnicities less accurately than do competing services from IBM and Microsoft."[44] In a 2018 study conducted by the ACLU, Rekognition incorrectly matched 28 members of Congress, identifying them as other people who have been arrested for a crime, further demonstrating how far this technology has to go, as well as the urgent need for Congress to act to pass effective laws and regulations on the market.[45]

B. U.S. Law vs. the Rest of the World and the EU's Gold Standard

a. The Current U.S. Regulatory Posture

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[41] *Id.*

[42] *We Are Implementing a One-Year Moratorium on Police Use of Rekognition*, AMAZON POL'Y NEWS & VIEWS (June 10, 2020), https://www.aboutamazon.com/news/policy-news-views/we-are-implementing-a-one-year-moratorium-on-police-use-of-rekognition.

[43] *Id.*

[44] Kyle Wiggers, *MIT Researchers: Amazon's Rekognition Shows Gender and Ethnic Bias*, VENTURE BEAT (Jan. 24, 2019, 7:50 PM), https://venturebeat.com/2019/01/24/amazon-rekognition-bias-mit/.

[45] Jacob Snow, *Amazon's Face Recognition Falsely Matched 28 Members of Congress with Mugshots*, ACLU BLOG (July 26, 2018), https://www.aclu.org/blog/privacy-technology/surveillance-technologies/amazons-face-recognition-falsely-matched-28.

Data Privacy laws govern facial recognition data collection, storage, and use. Facial recognition data falls under the category of biometric data specifically,[46] which is a special class that typically requires more protection than other types of data; a notion captured in many states' attempts to legislate protections for citizenry in light of the U.S. government's failure to enact a federal privacy law.[47] One example of these state laws is the Illinois Biometric Information Privacy Act ("BIPA"), which specifically notes that biometrics are "unlike other unique identifiers that are used to access finances or other sensitive information" in that they are "biologically unique to the individual; therefore, once compromised, the individual has no recourse, is at heightened risk for identity theft, and is likely to withdraw from biometric-facilitated transactions."[48] Yet, despite the clear need for proper regulatory frameworks to ensure the safe use of this technology, the concerning reality is that there are still many cities and states that do not offer comparable coverage and are struggling to keep up at the cost of their citizens' rights. The New York City Police Department ("NYPD"), for example, has demonstrated misuses of facial recognition technology during the Black Lives Matter ("BLM") movement. To illustrate, on August 7, 2020, dozens of NYPD officers equipped with riot gear appeared at the home of Derrick Ingram, a BLM activist, and refused to produce a warrant when Ingram requested it.[49] The NYPD found Ingram by using facial recognition technology, calling into question whether such use and subsequent showing of force was really necessary.[50] In yet another example, The Intercept reported that U.S. Marshals used

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[46] Thorin Klosowski, *Facial Recognition Is Everywhere. Here's What We Can Do About It*, N.Y. TIMES: WIRECUTTER (July 15, 2020), https://www.nytimes.com/wirecutter/blog/how-facial-recognition-works/.
[47] *See id*.
[48] 740 Ill. Comp. Stat. Ann. 14/5 (2008).
[49] James Vincent, *NYPD Used Facial Recognition to Track Down Black Lives Matter Activist*, THE VERGE (Aug. 18, 2020, 5:26 AM), https://www.theverge.com/2020/8/18/21373316/nypd-facial-recognition-black-lives-matter-activist-derrick-ingram.
[50] *Id.*

drones to spy on BLM protesters in Washington, D.C. during the summer of 2020, which underscored a "growing militarization of policing"[51] as well as the need for more regulatory scrutiny to be applied in such instances.

Unfortunately, redress for data privacy is difficult to obtain through the courts. One of the biggest reasons for this is that standing is notoriously difficult to establish in privacy cases.[52] There is no express cause of action for data privacy violations, nor is there an independent data protection authority committed to handling such redress. While the Federal Trade Commission ("FTC") aims to protect consumers "by stopping unfair, deceptive or fraudulent practices in the marketplace,"[53] the FTC typically engages in more large-scale initiatives and their resources limit their ability to handle the volume of charges that would be brought forth at the individual level. Therefore, individuals must typically rely on causes of action found in contracts, where specific provisions are carved out to protect consumers' data, torts, or criminal and constitutional theories relating to search and seizure to find a remedy.

In lieu of Congressional action, the common law has governed much of the adjudicative responsibility within the privacy space and has come a very long way in how it treats data privacy as a right in the U.S. Many years ago, in cases like *Olmstead*, the U.S. government and its courts agreed that The Fourth Amendment was not violated unless there had been an official search and seizure of a defendant's person, his papers or other tangible material effects.[54] If this rule were applied today, that would be problematic because one's effects are now largely intangible and less physical in nature, at least when it comes to data and general Internet presence. Historically, the courts' treatment of privacy has been progressively pro-data rights for individuals, and judicial interpretation has

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[51] Sam Biddle, *U.S. Marshals Used Drones to Spy on Black Lives Matter Protests in Washington, D.C.*, THE INTERCEPT (Apr. 22, 2021, 2:05 PM), https://theintercept.com/2021/04/22/drones-black-lives-matter-protests-marshals/.
[52] Margot E. Kaminski, *Standing After Snowden: Lessons on Privacy Harm from National Security Surveillance Litigation*, 66 DEPAUL L. REV. 413, 414 (2017).
[53] *What We Do*, FTC, https://www.ftc.gov/about-ftc/what-we-do (last visited Dec. 10, 2021).
[54] Olmstead v. United States, 277 U.S. 438, 466 (1928).

evolved substantially since *Olmstead*. *Katz* marked the shifting of the tide, wherein the courts started to acknowledge that the collection of less tangible data, via police eavesdropping, could still constitute a "search" under the Fourth Amendment, thus requiring a warrant.[55] Around the time of the *Katz* decision, the Court ruled in *Griswold v. Connecticut* that privacy is protected under the Constitution.[56] Specifically, the Court ruled that while this right is not explicitly contained in the text of the Constitution, the various guarantees within the Bill of Rights create "penumbras," or "zones," that establish a right to privacy.[57] Decisions like these paved the way for the modern common law interpretation of privacy rights, which now largely involve data rights issues.

Interestingly, data privacy issues in the common law do not align to a specific party's interests, and thus, tend to be bipartisan in nature, which is relatively rare in modern politics and highlights the degree of consensus around its importance. Though data privacy issues are typically associated with left-leaning institutions like the ACLU and the Electronic Frontier Foundation ("EFF"), major decisions that have been rendered by the Supreme Court in recent years have been by judges like the late Justice Antonin Scalia ("Scalia" or "Justice Scalia"), who was notoriously conservative.[58] One of the most influential Fourth Amendment privacy cases of the modern era was *Kyllo v. United States*, wherein a Department of the Interior agent used a high-tech thermal-imaging device to scan the interior of a criminal defendant's home to determine whether or not the suspect was growing marijuana.[59] Scalia, writing for the Court, found that obtaining information regarding the inside of a home, which could not otherwise have been obtained without physical intrusion into a constitutionally protected area, constituted a search,

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[55] Katz v. United States, 389 U.S. 347, 353 (1967).
[56] Griswold v. Connecticut, 381 U.S. 479, 484-85 (1965).
[57] *Id.*
[58] Adam Liptak, *Antonin Scalia, Justice on the Supreme Court, Dies at 79*, N.Y. TIMES (Feb. 13, 2016), https://www.nytimes.com/2016/02/14/us/antonin-scalia-death.html.
[59] Kyllo v. United States, 533 U.S. 27, 29 (2001).

at least where the technology was not in general public use.[60] This was significant because two important elements of the test relied on subjectivity and reasonableness. Specifically, Scalia focused on the idea that (1) the individual must manifest a subjective expectation of privacy and (2) society must be willing to recognize that expectation as reasonable,[61] which is a powerful test to apply to cases involving intangible, wrongfully obtained data by the government. Later, in *Jardines*, Scalia would also remove the requirement of the government's use of high-tech devices for the crux of *Kyllo* to be able to stand when police relied on the warrantless use of drug-sniffing dogs on a suspect's home.[62]

Despite these victories, the courts have acted slowly. It can take a long time and substantial resources for a case to reach the Supreme Court, and even then, because the Supreme Court takes cases on a discretionary basis, it can feel like achieving justice is on par with a kind of lottery system. Thus, relying on the judiciary to address critical issues of wrongfully obtained biometric data is not a truly viable solution.

To date, there is no firm judiciary consensus where biometric data collection is concerned, at least not at the Supreme Court level, where it could serve to inform national use of facial recognition technology. One of the closest and most relevant biometric data matters that made it to the Supreme Court was the 2013 case, *Maryland v. King*. In *King*, the specific biometric data at issue centered on the Maryland DNA Collection Act ("MDCA"), which permits local law enforcement to collect DNA samples from individuals arrested for violent crimes.[63] In that case, law enforcement used data in their DNA database to convict a criminal defendant of a separate crime from the original crime he committed where he first submitted that DNA.[64] In his dissent, Justice Scalia argued that the Fourth Amendment was meant to prevent the government from searching a person for evidence of a crime without

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[60] *Id.* at 34-35.
[61] *Id.*
[62] Florida v. Jardines, 569 U.S. 1, 11-12 (2013).
[63] Maryland v. King, 569 U.S. 435, 443 (2013).
[64] *Id.* at 441.

cause.[65] Scalia further pointed out that the decision in *King* would open the door for DNA tests to be conducted in the absence of evidence linking the arrestee to a specific DNA-related crime, which would constitute a "general warrant," which the Fourth Amendment was intended to prohibit.[66] Similarly, with facial recognition technology, the government essentially has total access to people's identities and whereabouts, which, while valuable for potential investigations, is problematic, especially where there is no reason to be surveilling average people. As of now, there simply is no definitive "standard for how you evaluate [facial recognition] and how you consider whether or not the technology makes sense for one application or another."[67]

b. Enter: GDPR & Maximilian Schrems

In May of 2018, the GDPR came into effect.[68] It would become one of the most impactful global regulations passed in modern times,[69] requiring compliance at a global scale in order for businesses to continue engaging in cross-border data exchanges with Europe.[70] Following its enactment, the impact was felt almost instantaneously as companies made substantial amendments to their respective privacy policies and engaged in comprehensive training for employees[71] at all levels to ensure cross-functional compliance. The GDPR's coverage is expansive, addressing concepts ranging from

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[65] *Id.* at 466 (Scalia, J., dissenting).
[66] *Id.* at 481.
[67] Ivanova, *supra* note 32, at 7.
[68] Joanna Kessler, Data Protection in The Wake of the GDPR: California's Solution for Protecting "The World's Most Valuable Resource," 93 S. CAL. L. REV. 99, 103 (2019).
[69] Matthew R. A. Heiman, *The GDPR And the Consequences of Big Regulation*, 47 PEPPERDINE L. REV. 945, 947 (2020).
[70] Council Regulation 2016/679, 2016 O.J. (L 119) 2 (EU) [hereinafter GDPR].
[71] Nancy Harris, *A Practical Guide to the European Union's GDPR for American Businesses*, VOX (May 16, 2018), https://www.vox.com/2018/5/16/17360944/gdpr-us-business-eu-european-union-data-protection-privacy.

privacy by design to standards of breach notification procedures.[72] However, the crux of the regulation can be summarized as bestowing greater control to consumers over their data while holding companies accountable for any misuses of the data they have access to.[73]

Despite the various updates to companies' privacy policies and other steps U.S. companies have taken towards complying with the GDPR, the U.S. continues to struggle to reassure the EU, and many consumers, of its integrity and overall efficacy through those efforts. The apprehension towards the U.S. is unsurprising given the frequency and magnitude of privacy controversies stemming from U.S. companies in recent years. Just in the short time between GDPR's effective date in 2018 and the present, the number of U.S. based mass-scale data breaches have been significant, including, *inter alia*:

- **SolarWinds**: Headquartered in Austin, TX, software company SolarWinds experienced one of the largest data breaches in history that spread to its clients and "went undetected for months."[74] Those impacted range from Microsoft, which had its source code compromised, to the U.S. government itself – the full extent of the breach is still being unraveled; while uncovered in 2020, investigations are ongoing as to when the breach began and whether it has been completely remediated.[75]

- **Twitter Celebrity Spearfishing Attack**: 2020 also saw what came to be known as the "worst-known hack of Twitter systems" when the accounts of public figures, such as Elon Musk, Jeff Bezos,

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[72] *See* Heiman, *supra* note 69.
[73] *See* Kessler, *supra* note 68.
[74] Isabella Jibillian & Katie Canales, *The US is Readying Sanctions Against Russia over the SolarWinds Cyber Attack. Here's a Simple Explanation of How the Massive Solarwinds Hack Happened and Why It's Such a Big Deal*, BUS. INSIDER, https://www.businessinsider.com/solarwinds-hack-explained-government-agencies-cyber-security-2020-12 (last updated Apr. 15, 2021, 1:25PM).
[75] *Id.*

Barack Obama, and Joe Biden were compromised in a bitcoin scam.[76] For this, hackers used a technique called "phone spear phishing" to gain access to upwards of 45 celebrity twitter accounts.[77]

- **Zoom**: As the pandemic-ridden world transitioned to a deeper reliance on virtual conferencing in 2020, Zoom was an increasingly adopted tele-meeting solution for corporate employees, teachers, and students alike who were all now working remotely. Unfortunately, hackers also saw this as a ripe opportunity and took full advantage, as it was discovered in April of 2020 that half a million Zoom passwords were found being sold on the dark web.[78] This was around the same time that Zoom found itself mired in controversy over a phenomenon known as "Zoombombing," whereby Internet trolls were engaging in unwanted, disruptive activity on Zoom calls they were not invited to.[79]

While it is unreasonable to fault companies for the activity of hackers, these breaches do beg questions around compliance with the spirit of regulations like GDPR. Have these companies properly secured the compromised data? For instance, did they establish proper controls for access?[80] Did they collect only the data necessary

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[76] Andy Greenberg, *The Attack That Broke Twitter Is Hitting Dozens of Companies*, WIRED (Aug. 18, 2020), https://www.wired.com/story/phone-spear-phishing-twitter-crime-wave/.
[77] *Id.*
[78] Davey Winder, *Zoom Gets Stuffed: Here's How Hackers Got Hold Of 500,000 Passwords*, FORBES (Apr. 28, 2020, 6:46 AM), https://www.forbes.com/sites/daveywinder/2020/04/28/zoom-gets-stuffed-heres-how-hackers-got-hold-of-500000-passwords/?sh=4fc961695cdc.
[79] Sara Morrison, *Zoom Responds to Its Privacy (And Porn) Problems*, VOX (Apr. 2, 2020, 11:44 AM), https://www.vox.com/recode/2020/3/31/21201019/zoom-coronavirus-privacy-hacks.
[80] *See* Heiman, *supra* note 69, at 2.

for them to be able to provide their intended service?[81] In its 2020 settlement with Zoom, the FTC addressed a series of issues but chief among them was that the company was allegedly misleading users about the privacy and security of its core product.[82] Namely, the director of the FTC's Bureau of Consumer Protection stated that "Zoom's security practices didn't line up with its promises."[83] This causes concern for consumers and also has the effect of furthering international discontent. One individual who has taken particular aim at the U.S. in recent years is Maximilian Schrems ("Schrems"), a data protection advocate who has applied significant scrutiny to tech giants like Facebook since he was a 23-year-old law student in Vienna.[84]

Admittedly, Schrems' issues with U.S. corporations are related more so to the U.S. government and its surveillance practices. Less than a year after GDPR took effect, Schrems sued Facebook Ireland Ltd. through the Irish Data Commissioner in 2011 over allegations relating to involvement between Facebook and the U.S.'s PRISM surveillance program.[85] PRISM, a code name for a National Security Agency ("NSA") program, started under the Bush administration and focuses on collecting stored internet communications based on demands made from the government to cross-industry companies, especially those within the technology sector, under Section 702 of the FISA Amendments to turn over any data that match certain court-approved search terms.[86] Its existence is only known due to Edward

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[81] *Id.*
[82] FTC Gives Final Approval to Settlement with Zoom over Allegations the Company Misled Consumers About Its Data Security Practices, FTC (Feb. 1, 2021), https://www.ftc.gov/news-events/press-releases/2021/02/ftc-gives-final-approval-settlement-zoom-over-allegations-company?utm_source=govdelivery.
[83] *FTC Requires Zoom to Enhance its Security Practices as Part of Settlement*, FTC (Nov. 9, 2020), https://www.ftc.gov/news-events/press-releases/2020/11/ftc-requires-zoom-enhance-its-security-practices-part-settlement.
[84] Hannah Kuchler, *Max Schrems: The Man Who Took on Facebook – And Won*, FIN. TIMES (Apr. 5, 2018), https://www.ft.com/content/86d1ce50-3799-11e8-8eee-e06bde01c544.
[85] Case C-362/14, Maximillian Schrems v. Data Prot. Comm'r, ECLI:EU:C:2015:650, ¶1 (Oct. 6, 2015).
[86] 50 U.S.C. § 1881a.

Snowden's infamous series of data leaks,[87] which also brings into question, for many foreign nations, and skeptics like Schrems, what other surveillance programs are currently being conducted by the U.S. The decision of the Court of Justice of the European Union ("CJEU") focused on the EU-U.S. Safe Harbor, which was a framework that was intended to prevent companies that store customer data within the EU or U.S. from accidentally disclosing or losing personal information and addressed whether or not the Safe Harbor provided adequate protection.[88] In its ruling, the CJEU sided with Schrems in determining that the Safe Harbor did not provide such adequate protection.[89] The CJEU found that (1) the Safe Harbor permitted unacceptable government interference, (2) did not provide legal remedies for individuals who sought to access data related to them or have it erased or amended, and (3) it prevented national supervisory authorities from exercising their powers.[90] Under EU law, cross-border data transactions with countries deemed to have lower, or "inadequate," privacy standards, including the U.S., are prohibited.[91] Following the ruling, the U.S. proposed a more robust alternative to the Safe Harbor via the EU-U.S. Privacy Shield, which was validated and adopted by the EU in July of 2016.[92]

In the summer of 2020, coinciding with the tensions in the U.S. surrounding the death of George Floyd, and on the almost 4-year anniversary of the EU-U.S. Privacy Shield, Schrems struck again when the CJEU struck the Privacy Shield down following the *Schrems II* decision.[93] Again, FISA and U.S. surveillance practices seemed to serve as the crux of the CJEU's decision.[94] Regarding the

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[87] Gianluca Mezzofiore, *NSA Whistleblower Edward Snowden: Washington Snoopers Are Criminals*, INT'L BUS. TIMES (July 1, 2014, 9:24 PM), https://www.ibtimes.co.uk/nsa-whistleblower-edward-snowden-479709.
[88] *Schrems,* ECLI:EU:C:2015:650 at ¶ 98.
[89] *Id.*
[90] *Id.* at ¶ 84.
[91] *See* GDPR, *supra* note 70, at 2.
[92] Europe Commission Press Release IP/16/2461, European Commission launches EU-U.S. Privacy Shield: Stronger Protection for Transatlantic Data Flows (July 12, 2016).
[93] Case C-311/18, ECLI:EU:C:2020:559 (July 16, 2020).
[94] *Id.*

validity of the European Commission's Privacy Shield Decision, the CJEU found that U.S. limitations on data protection violate the principle of proportionality, which requires that legislation must incorporate "clear and precise rules governing the scope and application of the measure in question and imposing minimum safeguards."[95] Moreover, any legislation infringing upon EU data subject's privacy rights must be "limited to what is strictly necessary."[96] The court ultimately concluded that U.S. limitations on data protection violate the principle of proportionality: "U.S. surveillance programs—specifically section 702 of FISA and Executive Order 12,333—do not impose "minimum safeguards" and are not "limited to what is strictly necessary."[97]

Despite the U.S.' best efforts and state-wide legislative attempts at remedying the absence of an effective, national privacy law (e.g., the California Consumer Privacy Act ("CCPA")[98] or BIPA), its current posture is best described as haphazard relative to the ever-growing number of international frameworks being promulgated. The U.S.' approach to privacy thus far has been fragmented and has proved difficult to track from a compliance standpoint. Companies can struggle to navigate state-specific laws, depending on where the companies operate, that are frequently being passed and do not always align with each other.

III. ANALYSIS

A. What's in a Face: The Public Policy Need to Protect Facial Image Data

a. HIPAA as Federal Acknowledgment of a Need to Protect Sensitive Biometric Data

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[95] *Id.*
[96] *Id.*
[97] Data Protection Commissioner v. Facebook Ireland Ltd.: *Court of Justice of the European Union Invalidates the EU-U.S. Privacy Shield*, 134 HARV. L. REV. 1567 (2021), https://harvardlawreview.org/2021/02/data-protection-commissioner-v-facebook-ireland-ltd/.
[98] CAL. CIV. CODE § 1798.110 (West 2020).

Despite there being no broadly applicable federal privacy law in the U.S., federal privacy laws do exist for the most precious and sensitive classes of data ascribed to individuals. One of the most well-known among these is the Health Insurance Portability and Accountability Act, more commonly referred to as "HIPAA."[99] Originally signed into law during the Clinton administration, the original intent of HIPAA was to "improve portability and continuity of health insurance coverage in the group and individual markets," combating health care fraud and abuse, "promot[ing] the use of medical savings accounts[,] . . . improv[ing] access to long-term care services and [insurance] coverage," and "simplify[ing] the administration of health insurance."[100]

Although the law was passed in a time when the extent of society's dependency on technology was based on a dial-up connection to America Online ("AOL"), and even aside from the amendments that would be passed through the subsequent administrations from Bush to Obama,[101] the heart of the law has always centered on patient privacy. This can be illustrated through what is known as the HIPAA Privacy Rule ("Privacy Rule"),[102] which "strives to balance the interest of individuals in maintaining the confidentiality of their health information with the interests of society in obtaining, using, and disclosing health information to carry out a variety of public and private activities."[103]

The control mechanisms of the Privacy Rule follow a general formula. HIPAA's "covered entities," health plans, health care clearinghouses, health care providers who electronically transmit any health information in connection with certain transactions, such as billing and payment for services or insurance coverage,[104] are

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[99] Health Insurance Portability and Accountability Act of 1996, Pub. L. No. 104-191, 110 Stat. 1936 [hereinafter HIPAA].
[100] Stacey A. Tovino, *A Timely Right to Privacy*, 104 IOWA L. REV. 1361, 1367-68 (2019).
[101] *Id.* at 1369.
[102] Privacy of Individually Identifiable Health Information, 45 C.F.R. §§ 164.500–164.534, Part 164, Subpart E (2016) [hereinafter Privacy Rule].
[103] *See* Tovino, *supra* note 100, at 1369.
[104] 45 C.F.R. § 160.103 (2017).

analogous to GDPR's data "controllers," such as " [a] natural or legal person, public authority, agency or other body which, alone or jointly with others, determines the purposes and means of the processing of personal data."[105] Similarly, HIPAA's "business associates," a person or entity conducting certain functions on behalf of a covered entity,[106] can be compared to GDPR's data "processors," a natural or legal person, public authority, agency or other body which processes personal data on behalf of the controller.[107] Additionally, the statute, its supplements, and its amendments all also conform to typical principles of privacy and security policy construction as it centers on the concepts of confidentiality, integrity, and availability.[108]

The very existence of HIPAA is itself an acknowledgement on the part of the U.S. that not only is biometric data, which, by definition, includes facial image data, highly sensitive but that it also requires proportionally heightened protection. Among its shortcomings, however, is that its scope narrowly targets entities like hospitals, academic medical research centers, and other health care providers. Because of this, commercial entities are not covered. Biotech and pharmaceutical companies exist in a space that is heavily regulated by the Food and Drug Administration ("FDA"), yet there are many instances of harmful circumvention that have led to catastrophes like opioid crises in the U.S.[109] Thus, relying on companies to do the right thing and follow the stewardship of companies like IBM and Amazon that have self-imposed pauses in the development of facial recognition technology is ill-advised. In short, absent the exercising of this kind of moratorium industry-wide, relying on tech companies to merely self-certify that they are taking the right steps is simply insufficient when dealing with this class of data.

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[105] *See* GDPR, *supra* note 70, at Art. 4, § 7 "Controller."
[106] 45 C.F.R.§§ 160.103.
[107] *See* GDPR, *supra* note 70, at Art. 4, § 8 "Processor."
[108] *See generally* 45 C.F.R. § 164(c) [hereinafter Security Rule].
[109] Patrick Malone et al., *Big Pharma's Twist: Billions in Opioid Settlements Convert into Huge Tax Breaks*, JD SUPRA (Feb. 18, 2021), https://www.jdsupra.com/legalnews/big-pharma-s-twist-billions-in-opioid-9993419/.

b. Other Statutory & Regulatory Examples of the Need to Federally Regulate Biometric Facial Data

HIPAA is not the only federal regulation that protects sensitive personal data. The finance industry, for example, is regulated by the Gramm-Leach-Bliley Act ("GLBA").[110] Similar to HIPAA and other privacy and security regulations, the GLBA also includes standard controls: it requires that covered financial entities (1) provide annual "privacy notices" that inform customers of their privacy practices, (2) permit consumers to prevent their personal information from being shared with non-affiliated companies (via an opt-out requirement), and (3) develop policies to promote data security.[111] Additionally, the GLBA creates a right of enforcement for federal agencies, including the FTC,[112] again, there is currently no private right of action for consumer data privacy violations in the U.S., however, a comprehensive privacy framework is not one that is built on fragmented, overly narrow approaches to data protection.

The goal for robust, effective regulation should always be focused on those who are responsible for compliance. To that end, the GDPR serves as an exemplar for this as its coverage is not industry specific. With regards to biometric data, the GDPR also treats this class of protected information as especially sensitive.[113] The GDPR defines biometric data as "personal data resulting from specific technical processing relating to the physical, physiological or behavioral characteristics of a natural person, which allow or confirm the unique identification of that natural person, such as *facial images* or dactyloscopic (fingerprint) data."[114] Former Senator Al Franken, in a 2012 congressional hearing on facial recognition, best underscored the importance of securing biometric facial recognition data when he said:

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[110] Gramm-Leach-Bliley Act of 1999, Pub. L. No. 106-102, 113 Stat. 1338 (1999) (codified in sections 12 and 15 U.S.C.) [hereinafter GLBA].
[111] *Id.*
[112] *Id.*
[113] *See* GDPR, *supra* note 70.
[114] *See id.* (emphasis added).

> I believe that we have a fundamental right to control our private information, and biometric information is already among the most sensitive of our private information, mainly because it is both unique and permanent. You can change your password. You can get a new credit card. But you cannot change your fingerprint, and you cannot change your face . . . the dimensions of our faces are unique to each of us . . . just like fingerprint analysis, facial recognition technology allows others to identify you with what is called a "faceprint"—a unique file describing your face. But facial recognition creates acute privacy concerns that finger- prints do not . . . . Your face is a conduit to an incredible amount of information about you, and facial recognition technology can allow others to access all of that information from a distance, without your knowledge, and in about as much time as it takes to snap a photo.[115]

c. The Issue with Digital Photography Itself

Concerns surrounding facial recognition technology are not just speculative; the shortcomings have, in fact, been proven.[116] Facial recognition developers tout high accuracy, but that is typically when the technology is tested in ideal environments that are conducive to near-perfect image quality.[117] However, surveillance applications do not offer such pristine conditions.[118] Often, facial images captured in

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[115] What Facial Recognition Technology Means for Privacy and Civil Liberties: Hearing Before the Subcomm. on Priv. Tech. and the L. of the Comm. on the Judiciary U.S. S., 112th Cong. 1-2 (2012) (statement of Hon. Al Franken, U.S. Sen. of Minn.).
[116] *See supra* note 16, Section II.
[117] *See* GROTHER ET AL., *supra* note 26, at 4.
[118] Jennifer Valentino-DeVries, *How the Police Use Facial Recognition, and Where It Falls Short*, N.Y. TIMES (Jan. 12, 2020), https://www.nytimes.com/2020/01/12/technology/facial-recognition-police.html.

surveillance footage are captured in dim lighting, contain faces turned at an angle, or are distorted with the use of face masks or even "with minimal disguises such as baseball caps or sunglasses hamper[ing] accuracy."[119] Thus, while we have advanced far beyond the technological capabilities of Bledsoe's era, challenges still linger. This is exacerbated by the fact that there is a racial bias built into digital photography itself, even aside from embedded algorithms and facial recognition technology specifically.[120] This is why people of darker skin tones have trouble being captured on Zoom calls, recognized in Snapchat filters, or are instructed not to wear clothing of lighter colors than their faces when being recorded.[121] The problem of unconscious bias has seeped into photography for almost as long as color photography has existed:

> Photography is not just a system of calibrating light, but a technology of subjective decisions. Light skin became the chemical baseline for film technology, fulfilling the needs of its target dominant market. For example, developing color-film technology initially required what was called a Shirley card. When you sent off your film to get developed, lab technicians would use the image of a white woman with brown hair named Shirley as the measuring stick against which they calibrated the colors. Quality control meant ensuring that Shirley's face looked good. It has translated into the color-balancing of digital technology.[122]

Therefore, in the police's fight for justice, it is worth questioning the validity and reliability of arrest outcomes that use facial

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[119] *See generally id.*
[120] Sarah Lewis, *The Racial Bias Built into Photography*, N.Y. TIMES (Apr. 25, 2019), https://www.nytimes.com/ 2019/04/25/lens/sarah-lewis-racial-bias-photography.html.
[121] *See, e.g.*, *id.*
[122] *Id.*

recognition technology when it targets certain groups of people or has been proven to demonstrate favoritism towards others. Based on current data, and given its current applications, the U.S.' failure to federally regulate facial recognition technology is almost negligent and demonstrates a harmful indifference towards those who will be prosecuted disproportionately as a function of the continued use of the technology by these agencies.

B. The Business Case for Passing a Federal Biometric Data Law

a. The Impact of Privacy in Corporate Culture

In addition to unconscious bias, there are also economic incentives for creating federally recognized standards. Corporate practices are becoming commensurately more scrutinized internally and externally. Following the death of George Floyd, the global gaze was turned on the President of the United States and, unfortunately, many found the president's responses to be either underwhelming or, in some instances, downright harmful. One example of this was the Facebook walkout of 2020.[123] In response to President Trump's inflammatory Facebook posts and Facebook's failure to intervene or provide a meaningful response, hundreds of employees conducted a "virtual walkout," in light of the quarantine.[124] These employees chose personal values and principles over the threat of retaliation or loss of production, with one such demonstrator posting on an internal message board: "The hateful rhetoric advocating violence against black demonstrators by the US President does not warrant defense under the guise of freedom of expression."[125] This was similar to the Google work stoppage of 2018, when Claire Stapleton, a YouTube employee, and 20,000 or so of her co-workers (globally), walked out

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[123] *See*, *e.g.*, Sheera Frenkel et al., *Facebook Employees Stage Virtual Walkout to Protest Trump Posts*, N.Y. TIMES (June 1, 2020), https://www.nytimes.com/2020/06/01/technology/facebook-employee-protest-trump.html.
[124] *Id.*
[125] *Id.*

of their offices in protest of Google's $90 million payout to a former executive accused of credible sexual harassment claims.[126] Among their requests was an end to mandatory arbitration and a public sexual harassment transparency report.[127] From this, I deduce the following: (1) employees are increasingly less fearful of taking risks to ensure that the companies they work for align with their principles and values and (2) since work is increasingly remote, especially following the pandemic, it is easier to organize digitally and to influence employees at a global scale – protest is always potentially one Slack message away.

The cost of a work stoppage for thousands of employees is substantial, and so from a policy perspective, tech companies have it in their best interest to conform their practices with modern social trends and perspectives on human rights. Per *Griswold*,[128] and the relatively recent wave of data-conscientiousness among consumers globally,[129] data privacy, especially the particularly sensitive classes (i.e., biometric/facial data), is one of those rights that will need to be built into corporate culture to ensure that productivity and morale are not impacted.[130] Indeed, the impact of corporate culture, employee morale, and the relationship between the two is empirically proven.[131] For example, increased diversity, of views and background, not just racial or ethnic, and reduced cultural conflict have been shown to enhance innovation and produce higher quality

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[126] Johana Bhuiyan, *How the Google Walkout Transformed Tech Workers into Activists*, L.A. TIMES (Nov. 6, 2019), https://www.latimes.com/business/technology/story/2019-11-06/google-employee-walkout-tech-industry-activism.
[127] *Id.*
[128] Katz v. United States, 389 U.S. 347 (1967) (referencing the Court's recognition of the individual's right to Privacy as a Constitutional right).
[129] Thomas C. Redman & Robert M. Waitman, *Do You Care About Privacy as Much as Your Customers Do?*, HARV. BUS. REV. (Jan. 28, 2020), https://hbr.org/2020/01/do-you-care-about-privacy-as-much-as-your-customers-do.
[130] Gary B. Gorton & Alexander K. Zentefis, *Social Progress and Corporate Culture* (Feb. 26, 2021) (Nat'l Bureau of Econ. Research, Working Paper No. 25484) (on file with the Yale School of Management).
[131] *Id.*

decisions, yielding overall better financial performance for firms.[132] This is partially why the concept of "Privacy by Design," as referenced in the GDPR,[133] is a recent and sought after cultural component by many modern, progressive tech firms today. The other reason is because it is a concept embraced by most regulators and offers a broader, streamlined approach to privacy compliance,[134] which translates to reduced risk exposure for having hefty privacy violation fines levied against those who adopt and practice the principles of privacy by design.[135] In simple terms, privacy by design calls for privacy being ingrained into everything that corporations develop and that impacts its employees, consumers, and society as a whole. Whether it is a product offering or a policy, privacy must remain top of mind to ensure that individuals are afforded the protections they are entitled to.

b. The Costs of Failing to Respect Privacy in the Marketplace

Privacy by design may be an ambitious ideal and not necessarily required to achieve compliance and strong corporate cultures, however, a general respect for privacy is at least a critical baseline when contemplating potential product development initiatives. If companies fail to seriously consider the privacy implications of their offerings' possible applications, they risk incurring considerable costs both in terms of direct financial losses and loss of consumer confidence. For instance, Illinois has been wielding BIPA against non-compliant companies since its passage in 2008, as it offers a private right of action for Illinois citizens for biometric privacy

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[132] *Id.*
[133] *See* Heiman, *supra* note 69.
[134] Ira S. Rubinstein & Nathaniel Good, Privacy by Design: A Counterfactual Analysis of Google and Facebook Privacy Incidents, 28 BERKELEY TECH. L.J. 1333, 1335 (2013).
[135] *Id.* at 1337 (discussing the seven principles of privacy by design, i.e., (1) Proactive not Reactive; Preventative not Remedial; (2) Privacy as the Default Setting; (3) Privacy Embedded into Design; (4) Full Functionality—Positive-Sum, not Zero-Sum; (5) End-to-End Security—Full Lifecycle Protection; (6) Visibility and Transparency—Keep it Open; and (7) Respect for User Privacy).

violations and can result in costs of up to $5,000 per violation.[136] In February of 2020, Facebook settled a class-action BIPA lawsuit for $650 million,[137] which is among the highest amounts a company has ever had to pay for a privacy violation in a private lawsuit.[138] In that case, Facebook users targeted one of the most violative aspects of biometric facial data collection: lack of consent.[139] Through its "Tag Suggestions" tool, Facebook leveraged facial recognition software to identify the faces of users in images uploaded on the platform without their express consent.[140] While achieving express consent for nearly three billion Facebook users[141] may seem impractical, two things should be noted: (1) the impracticality of seeming tediousness is not a legal defense to the requirement of consent and (2) companies are able to achieve high volume consent through simple opt-in's (e.g., having users actively tick a box to express consent via a digital form). In fact, Facebook did eventually change its privacy policy to require opt-in only use of facial recognition on its platform roughly four years following the filing of this action in 2019.[142]

Unfortunately for Facebook, it has, in many ways, served as a precedent-setting cautionary tale to many companies – and not just within the tech industry alone, as the practice of consumer data collection spans almost all industries, especially for companies that offer ecommerce solutions. Facebook currently holds the record for having the highest penalty in a privacy enforcement action by the

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[136] 740 Ill. Comp. Stat. Ann. 14/20 (2008).
[137] Patel v. Facebook, Inc., 932 F.3d 1264 (9th Cir. 2019), cert. denied, 140 S. Ct. 937 (2020).
[138] Taylor Hatmaker, *Facebook Will Pay $650 Million To Settle Class Action Suit Centered on Illinois Privacy Law*, TECHCRUNCH (Mar. 1, 2021), https://techcrunch.com/2021/03/01/facebook-illinois-class-action-bipa/.
[139] *See Patel*, 932 F.3d at 1273.
[140] *See id.* at 1268.
[141] *Number of Monthly Active Facebook Users Worldwide as of 3rd Quarter 2021*, STATISTA (Oct. 10, 2021), https://www.statista.com/statistics/264810/number-of-monthly-active-facebook-users-worldwide/.
[142] Kim Lyons, *Judge Approves $650 Million Facebook Privacy Settlement over Facial Recognition Feature*, THE VERGE (Feb. 27, 2021, 12:09 PM), https://www.theverge.com/2021/2/27/22304618/judge-approves-facebook-privacy-settlement-illinois-facial-recognition.

FTC for its involvement in the Cambridge Analytica scandal, in the amount of $5 billion.[143] For scale, this represents more than eleven times the penalty Equifax incurred when cybercriminals accessed over 145 million consumers' sensitive personal data, including full names, social security numbers, birth dates, addresses, and driver license numbers.[144] For its part in the scandal, ". . . Facebook had allowed a data firm named Cambridge Analytica to harvest user data without permission from more than 50 million user profiles (later estimated to be as many as 87 million users) for controversial use by a campaign during the 2016 United States presidential election."[145] Among the many reasons for the large penalty were: one, it was revealed that Facebook was aware of—and permitted—Cambridge Analytica's pillaging of consumer data.[146] Two, though it had the power to intervene prior to charges being filed, Facebook failed to do so.[147] Three, Facebook's involvement directly contradicted its privacy policies, wherein it promised users that they had control over their data—a promise ultimately significantly undermined.[148] And finally, the FTC found that "Facebook monetizes user information through targeted advertising, which generated most of the company's $55.8 billion in revenues in 2018."[149] That last point is significant because it demonstrates both the extent of some companies' reliance on these kinds of data collection practices from a revenue standpoint as well as the risk magnitude from a potential

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[143] *FTC Imposes $5 Billion Penalty and Sweeping New Privacy Restrictions on Facebook*, FTC (July 24, 2019), https://www.ftc.gov/news-events/press-releases/2019/07/ftc-imposes-5-billion-penalty-sweeping-new-privacy-restrictions.
[144] *Equifax Data Breach Settlement*, FTC, https://www.ftc.gov/enforcement/cases-proceedings/refunds/equifax-data-breach-settlement (last visited Dec. 11, 2021); *See* Brian Fung, *145 Million Social Security Numbers, 99 Million Addresses and More: Every Type of Personal Data Equifax Lost to Hackers, By the Numbers*, WASH. POST (May 8, 2018), https://www.washingtonpost.com/news/the-switch/wp/2018/05/08/every-type-of-personal-data-equifax-lost-to-hackers-by-the-numbers/.
[145] Joseph T. McClure, A New Trend in Securities Fraud: Punishing People Who Do Bad Things, 48 TEX. J. BUS. L. 28, 42 (2020).
[146] *Id.*
[147] *Id.* at 42-43.
[148] FTC, *supra* note 143.
[149] *Id.*

penalty for that same reliance. The lesson from Cambridge Analytica translates to a need for companies to embrace principles of trust and transparency but to also make sure to follow through on its policies and promises and to always put the rights and values of their consumers first.

Of course, Facebook is not the only company to face legal consequences in this ineffectively regulated environment. Perhaps the best example of harmful indifference towards the rights of individuals is Clearview AI, which has faced lawsuits across at least four states for unethical practices related to the use of facial data collection.[150] According to the ACLU, amongst the numerous civil claims filed against Clearview AI across Vermont, Illinois, New York, and California, is its offering of a "massive faceprint database to private companies, police, federal agencies, and wealthy individuals, allowing them to secretly track and target whomever they wished using face recognition technology."[151] What sets Clearview AI apart from its competitors is not only the accuracy and efficacy of its tool in correctly identifying individuals' identities but the means by which it "scrapes" the image data from social media sites, which has resulted in a database of over 640 million images.[152] It is this tactic, among other things, that has drawn the ire of activists, private citizens, the ACLU, and even the social media sites themselves—many of which have been forced to issue cease and desist letters against the company for doing so.[153] To further

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[150] Nick Statt, *Clearview AI to Stop Selling Controversial Facial Recognition App to Private Companies*, THE VERGE (May 7, 2020, 8:29 PM), https://www.theverge.com/2020/5/7/21251387/clearview-ai-law-enforcement-police-facial-recognition-illinois-privacy-law.

[151] Nick Statt, *ACLU Sues Facial Recognition Firm Clearview AI, Calling It a 'Nightmare Scenario' for Privacy*, THE VERGE (May 28, 2020, 1:13 PM), https://www.theverge.com/2020/5/28/21273388/aclu-clearview-ai-lawsuit-facial-recognition-database-illinois-biometric-laws.

[152] *See Face Recognition Technology: DOJ and FBI Have Taken Some Actions in Response to GAO Recommendations to Ensure Privacy and Accuracy, But Additional Work Remains*: *Hearing on GAO-19-579T Before the H. Comm. on Oversight & Reform*, 116th Cong. 2n. 5 (2019) (statement of Gretta L. Goodwin, Director of Homeland Security and Justice) [hereinafter Goodwin Testimony].

[153] *See* Statt, *supra* note 150.

contextualize the scale of this impropriety, the Facebook tag suggestions example also dealt with the use of photo collection to enhance its tool and the lack of consent for Facebook users who uploaded those photos for other purposes.[154] The critical difference here is the scale of data collection without consent on the part of Clearview AI. Clearview AI engaged in the same practice except it collected presumably a much larger data set across Facebook, LinkedIn, and other similar platforms where users post photos.[155] This is further problematic because basic data privacy frameworks often require, at a minimum, what is called a "purpose limitation," a term defined in the GDPR.[156] This means that companies must pay attention to the purpose for which users are providing data and only process that data in a way that conforms to their expectations. It would be very difficult for Clearview AI to attempt to argue that users reasonably expected that their data would be used in the way it is using it, especially considering that most of those "users" are hardly that, as they never even uploaded that data to Clearview AI's databases directly nor are they otherwise formally associated with the company.

In today's landscape, companies therefore face the dual-cost risks associated with private litigation as well as fines from a multitude of governing authorities. The perpetrator of the largest GDPR violation, in terms of cost, to date is Amazon, which has incurred a fine of approximately $888 million.[157] Amazon, like Facebook and many other violators over the years, has taken the position that its collection of user data has been to improve the customer experience but the fine issued by the Luxembourg data

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[154] *See generally* Patel v. Facebook, Inc., 932 F.3d 1264 (9th Cir. 2019), cert. denied, 140 S. Ct. 937 (2020).

[155] Jon Porter, *Facebook and LinkedIn Are Latest to Demand Clearview Stop Scraping Images for Facial Recognition Tech*, THE VERGE (Feb. 6, 2020, 3:22 PM), https://www.theverge.com/2020/2/6/21126063/facebook-clearview-ai-image-scraping-facial-recognition-database-terms-of-service-twitter-youtube.

[156] *See* Heiman, *supra* note 69, at 5-6, 12, 25, 30.

[157] Stephanie Bodoni, *Amazon Gets Record $888 Million EU Fine Over Data Violations*, BLOOMBERG (July 30, 2021, 7:43 AM), https://www.bloomberg.com/news/articles/2021-07-30/amazon-given-record-888-million-eu-fine-for-data-privacy-breach.

protection authority has made it clear that such reasoning is grossly insufficient.[158] Amazon has engaged in a number of questionable privacy practices over the years—specific to biometric data has been the collection of consumer palm prints, via Amazon One, which allows consumers to pay for goods by swiping their hands.[159]

For now, some defenses available to combat evidence procured through facial recognition technology range from objecting under the *Daubert*[160] standard to citing a violation of the Confrontation Clause.[161] However, these defenses all have substantial limitations. First, under *Daubert*, prosecutors can still use the fruits of the evidence obtained (e.g., using extrinsic evidence, such as testimony from witnesses identified through investigation of the identified defendant and avoiding the challenge altogether) or rely on the studies that demonstrate the high accuracy of the technology.[162] Second, while the Confrontation Clause grants defendants certain rights to face their accusers,[163] including the use of hearsay evidence without an opportunity for the defendant to cross-examine the declarant,[164] facial recognition technology is unlikely to rise to the level of being challengeable. This is because the evidence does not constitute hearsay, as the technology and its analysts are not determining culpability, they are merely objectively identifying a

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[158] *Id.*

[159] Zack Whittaker, *Amazon Will Pay You $10 In Credit for Your Palm Print Biometrics*, TECH CRUNCH (Aug. 2, 2021, 2:49 PM), https://techcrunch.com/2021/08/02/amazon-credit-palm-biometrics/.

[160] Daubert v. Merrell Dow Pharms., Inc., 509 U.S. 579, 587, 595 (1993) (asking whether (1) the technique or theory has been tested; (2) it has been subjected to peer review; (3) its error rate was known; (4) standards for its operation existed and were maintained; and (5) it was generally accepted in the scientific community. *Daubert* placed the burden of proof on the proponent of the evidence. In 2000, drafters and the Court amended the Federal Rules of Evidence to incorporate *Daubert* factors as Rule 702.99).

[161] U.S. CONST. amend. VI.

[162] Ari B. Rubin, A Facial Challenge: Facial Recognition Technology and the Carpenter Doctrine, 27 RICH. J.L. & TECH. 3, 25 (2021).

[163] U.S. CONST. amend. VI.

[164] Crawford v. Washington, 541 U.S. 36, 68 (2004).

person.[165] Additionally, given its accuracy and the increased standardization and use of the technology, it may end up being used and regarded similarly to tools like police radar guns.[166]

Among the many reasons that companies collect user data is to offer their customers a better, more tailored user experience,[167] which is mutually beneficial because a better user experience increases the likelihood that a customer will return—a common sense concept in retail that long predates the Internet. In the example of Facebook's tag suggestions tool, the intent to improve the user experience is clear but as valuable as user experience is in theory, as a matter of public policy, it does not surpass the value of consumers' rights to their data and their being able to have a say in how their data is used and collected.[168] Companies would be better served by a federal, standardized data privacy law with robust enforcement mechanisms, rather than merely self-certifying their compliance, which can lead to massive financial losses, as seen in the Facebook-Cambridge Analytica settlement. These companies do not necessarily have nefarious intent to do harm, but innovations that intersect with human rights can lead to complex issues. For additional perspective, the Facebook examples, as problematic as they are, were related to behavioral trends, known in the aggregate as "psychographics," —it targeted the collection of such data points as the content users chose to post and clicks they made while using that platform.[169] Biometric facial data, in contrast, is not something that individuals volunteer, it is a literal part of them and gaining the needed consent for high volume surveillance and facial scanning is difficult, unlikely, and infeasible. As such, one could surmise the

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[165] *See* Williams v. Illinois, 567 U.S. 50, 86 (2012).
[166] *See* Rubin, *supra* note 162, at 42.
[167] Max Freedman, *How Businesses Are Collecting Data (And What They're Doing with It)*, BUS. NEWS DAILY (June 17, 2020), https://www.businessnewsdaily.com/10625-businesses-collecting-data.html.
[168] Tony Raval, *Data Privacy as A Basic Human Right*, FORBES (Nov. 12, 2019, 9:00 AM), https://www.forbes.com/sites/forbestechcouncil/2019/11/12/data-privacy-as-a-basic-human-right/?sh=40f3e2a24cbf.
[169] Sue Halpern, *Cambridge Analytica and the Perils of Psychographics*, THE NEW YORKER (Mar. 30, 2018), https://www.newyorker.com/news/news-desk/cambridge-analytica-and-the-perils-of-psychographics.

possible value of this kind of data and the proportional penalty a company be liable for when engaging in improper conduct centering on this type of data collection.

Lastly, if the U.S. fails to substantially curb its current surveillance practices, it will put itself in a precarious position in the global marketplace. In November of 2020, the European Data Protection Board ("EDPB"), issued a set of guidelines and recommendations for companies to abide by in order to fully comply with the content and spirit of the GDPR, as well as avoid the risk of losing the ability to engage in cross-border data transfers.[170] Attorney Brian G. Cesaratto effectively summarized the CJEU's decision in *Schrems II* and the ultimate impetus for the guidelines:

> [T]he outcome in *Schrems II* invalidating the Privacy Shield because it did not provide an adequate level of protection, was motivated by the Court's concern that the United States' surveillance programs broadly permitted surveillance of communications in transit for the objective of obtaining foreign intelligence without guarantees for non-U.S. persons potentially impacted by those programs, including the lack of actionable legal remedies against U.S. authorities . . . the Court recognized that these same concerns over broad surveillance programs warranted supplemental protections to protect the data.[171]

Through this lens, it is clear that current surveillance practices in the U.S., via the many privately developed and distributed facial recognition technology solutions for government use, poses a significant risk for compliance. Among the numerous

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[170] Brian G. Cesaratto, *The European Data Protection Board Issues Guidance on Cross-Border Data Transfers*, NAT'L L. REV. (Nov. 12, 2020), https://www.natlawreview.com/article/european-data-protection-board-issues-guidance-cross-border-data-transfers.
[171] *Id.*

recommendations offered in the EDPB's guidance is the requirement that organizations conduct a comprehensive assessment of "third countries," (i.e., countries other than EU member states and the three additional EEA countries including Norway, Iceland, and Liechtenstein) that have adopted a national law implementing the GDPR. This assessment of "effectiveness" is premised on whether the third country affords the data a level of protection equivalent to that provided by the GDPR. This step may require the parties to analyze whether such laws actually impede the effectiveness of the safeguards by evaluating "whether surveillance measures . . . by public authorities in a third country . . . can be regarded as a justifiable interference or not."[172] As an example, police, and other government, partnerships with companies like Clearview AI put a strain on EU regulators' ability to apply this kind of a requirement favorably. Thus, the ability to effectively comply with the GDPR is in conflict with the continuance of such practices and deregulation of facial recognition technology. While this presents potential risks of penalty-related costs associated with the GDPR, another penalty would be the limitation on U.S. companies' ability to engage in cross-border data transfers, which inevitably puts a freeze on global business interactions. This is because many modern businesses leverage ecommerce and Cloud technologies, which over the years have contributed to an increasingly globalized economy, and which rely on these types of cross-border transfers.

c. The Marketplace Benefits of Embracing Strong Data Privacy Policies

Companies should not develop robust, enforceable data privacy policies merely out of fear of lawsuits and FTC penalties. Instead, companies should embrace the positive outcomes that come along with pioneering positive change in this space. As consumers become

[172] Aaron K. Tantleff et al., *European Data Protection Board Issues Recommendations for Exports of Personal Data from the European Economic Area*, FOLEY & LARDNER LLP: PRIV., CYBERSECURITY & TECH. L. PERSP. (Dec. 1, 2020), https://www.foley.com/en/insights/publications/2020/12/european-data-protection-board-recommendations.

increasingly data conscious, companies have begun publishing publicly accessible documents like law enforcement transparency reports, for example, as a way to proactively communicate the frequency, types, and extent of government requests for consumer data.[173] Pioneered by Google in 2010, the concept of the transparency report has become an industry norm and has been adopted by every tech company in the Fortune 500.[174] However, the decision to publish these kinds of materials did not occur as a result of companies wanting to conform with Google, necessarily; rather, companies began to take on this responsibility after Edward Snowden revealed the massive surveillance conducted by the NSA, in 2013.[175] Regardless of whether companies sided with Snowden, they had to accept the new social consciousness that he helped to create. First, consumers became far more aware of the value of their privacy and the threat the government and other entities could pose to it, and companies have an economic interest in catering to the expectations of their consumers.[176] Secondly, at a minimum, the "European Parliament characterized Snowden as a hero for revealing the spying against Germany"[177]—this is the same European Parliament that passed the GDPR, which U.S. companies and states alike have accepted as the gold standard for privacy regulation.[178] Therefore, companies have an interest in rising to the expectations of its customer base in the U.S. and in the EU. This translates to an opportunity in the marketplace for gaining a competitive advantage through privacy-forward branding, thus the publication of transparency reports and other similar initiatives serve to benefit

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[173] Dashveenjit Kaur, *Amazon Shared Users' Data More Than 30,000 Times In 2020*, TECH HQ (Feb. 8, 2021), https://techhq.com/2021/02/amazon-shared-users-data-more-than-30000-times-in-2020/.
[174] *Id.*
[175] Jon M. Garon, *Seeking to Promote Security over Privacy and Achieving Neither*, 19 SMU SCI. & TECH. L. REV. 351, 358-59 (2016).
[176] *See* Freedman, *supra* note 167.
[177] *See* Garon, *supra* note 175, at 360.
[178] Isabella Anderson, Note, The Threat of Data Misuse as an Injury-in-Fact: Establishing A Uniform Framework for Constitutional Standing in the Privacy Era, 123 W. VA. L. REV. 263, 282 (2020).

companies while simultaneously advancing principles of trust and transparency.

Still, while the market has proven some ability to self-regulate, the government must step in to codify these customs. The Fortune 500 is not representative of the tech space in its entirety and while companies like IBM and Apple have established themselves as pioneers in the space of privacy-forward policy, it bears mentioning that many of the companies responsible for the current development and sale of facial recognition technologies are outside of the Fortune 500[179] and, thus, typically outside of the scope of these norms; very few, if any, make such an effort towards transparency. Size and salience should not be the catalysts for accountability, instead, the focus should be on the impact potential of these kinds of technologies—impact to our society and to our collective welfare through the risks posed to our most sensitive data.

C. Approaches to Effective Biometric Data Regulations

a. How the FAA Approached Drone Regulation and Why the U.S. Government Should Do the Same for Facial Recognition Technology

Prior to the mainstream coverage and concerns surrounding facial recognition technology, drones were the equivalent discussion point intersecting law, business, and public policy. Before their massive popularization, drones were heavily regulated by the Federal Aviation Administration ("FAA"). In fact, the FAA issued a conditional ban on the use of commercial drones outright in 2007, whereby such use was only allowed with special permission from the FAA.[180] From a safety perspective, the FAA had to take into account the fact that drones present the possibility of collisions with helicopter rotors or airplane engines or could even crash into and

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[179] Facial Recognition Companies, CRUNCHBASE, https://www.crunchbase.com/hub/facial-recognition-companies (last visited Dec. 1, 2021).

[180] Richard C. Balough, *Under Current Law, There's No Place for Commercial Drones*, CBA REC., Apr.-May 2015, at 34, 35.

harm people.[181] There is also the looming threat of a potential terrorist attacks. For example, drones are already used internationally in combat situations since they are capable of delivering and deploying explosives.[182] Additionally, there is less risk of capture or physical harm to the wrongdoer since the pilot is physically located a variable, and potentially substantial, distance away.[183] Lastly, drones are a cost-effective alternative to procuring other means of aircraft or vehicle with similar capability.[184] For these reasons, it took until about half a decade after the issuance of that commercial ban for Congress to pass the 2012 Modernization & Reform Act.[185] Yet, even then, the act called for the development of a five-year roadmap for introducing drones into the National Airspace System ("NAS") "to initiate a rulemaking on small unmanned aircraft."[186]

In lieu of a federal agency, or law, dedicated to the regulation of data privacy, many states have been forced to do what the FAA, and companies like IBM and Amazon, did: ban the use of facial recognition technology until effective regulations have been put in place to minimize potential negative impacts to society, or at least until proper assessments into the impact potential have been conducted, and proportional mitigations strategies have been proposed. These states include Washington, California, New York, and Massachusetts.[187] While the imminent threat that facial recognition technology poses is less obvious than a possible terrorist attack, the current unregulated use of the technology does erode many human and constitutional rights. Indeed, the motivation for the imposed bans center on the fact that present applications put "our

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[181] *Id.*
[182] Tung Yin, *Game of Drones: Defending Against Drone Terrorism*, 2 TEX. A&M L. REV. 635, 644 (2015).
[183] *Id.* at 645.
[184] *Id.* at 646.
[185] FAA Modernization & Reform Act of 2012, Pub. L. No. 112-95, 126 Stat. 11 (2012).
[186] *See* Balough, *supra* note 180, at 35.
[187] Vivian D. Wesson, *Why Facial Recognition Technology Is Flawed*, 92 N.Y. ST. B.J. 20, 21 (2020).

democratic freedoms and our civil liberties at risk."[188] Among other things, facial recognition technology has been recognized as creating a chilling effect that mass surveillance has on peaceful protests,[189] as seen through the demonstrations across the U.S. in the summer of 2020, calling for police reform after the death of George Floyd.[190]

Because of (1) the demonstrated bias embedded into facial recognition technology against minority groups, (2) the threat it poses to U.S. constitutional rights, and (3) the fact that it collects the most precious, sensitive data, any harm stemming from which is irrevocable due to its nature, there is simply no legitimate reason why the self-imposed moratorium companies and states have advanced should not be extended at a national, industry-wide level.

b. The California Model: CCPA, CPRA, and CPPA as a Framework for Federal Policy

An effective, robust, comprehensive federal law can preempt and remediate much of the potential harm of the more questionable and unethical facets of facial recognition technology. In drafting such a regulation, its construction would need to conform to the framework of the GDPR. Failure to do so would be counterproductive, as one goal of such a law should be to ensure broadly covered compliance to avoid limitation and constraints on cross-border data transfers. Thus, it would likely mirror other statutory frameworks that were based on the GDPR, such as the CCPA.[191] Being enacted only two years following the effective date of GDPR in 2018, the CCPA "strikingly mirrors the GDPR in several ways, making it the most notable state data privacy legislation."[192] However, the CCPA falls short in its failure to appropriately address

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[188] WASH. REV. CODE § 43.386.900 (effective July 1, 2021).
[189] *See* Wesson, *supra* note 187.
[190] Evan Selinger & Albert F. Cahn, *Did You Protest Recently? Your Face Might Be in A Database*, THE GUARDIAN (July 17, 2020), https://www.theguardian.com/commentisfree/2020/jul/17/protest-black-lives-matter-database.
[191] *See* CAL. CIV. CODE § 1798.100 (West 2018).
[192] Elizabeth L. Feld, United States Data Privacy Law: The Domino Effect After the GDPR, 24 N.C. BANKING INST. 481, 490 (2020).

the importance of biometric data—while it defines biometric data, it does not substantively contain provisions specific to biometric data protection beyond that.[193]

California appears to have attempted a course-correction in a 2020 ballot initiative, entitled the California Privacy Rights Act of 2020 ("CPRA"), that was adopted by voters in November of that year and set to take effect as of January of 2023.[194] One critical change in the CPRA (also referred to as the "CCPA 2.0") was taking a step closer to conformity with the GDPR by pulling biometric data into a category of especially sensitive data.[195] It even goes so far as to empower California consumers to limit use and disclosure of this sensitive personal information ("SPI") by requesting as much from companies, as well as increasing transparency overall by imposing specific disclosure requirements on businesses related to this kind of information.[196] Another critical provision captured by the CPRA is special enforcement rights. Rather than having to depend on the FTC to penalize companies, the CPRA empowers California citizens to pursue private actions through a newly created regulatory agency known as the California Privacy Protection Agency ("CPPA").[197] This is significant because it represents a rather toothy enhancement to the CCPA and also takes California and California businesses a step closer towards securing U.S. business relationships with the EU and the broader International economy, which has also started to take substantial steps towards conforming with the GDPR (e.g., Brazil, via the "LGPD," China via the "PDPL," India via the "PDPB," and many others).[198] Thus, it is likely that a federal privacy law would enhance and standardize much of the substance of the CPRA. Indeed, a national data protection agency ("DPA") is needed to

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[193] CAL. CIV. CODE § 1798.140(b) (West 2018).
[194] CAL. CIV. CODE § 1798.100, *amended by* California Privacy Rights Act of 2020, sec. 2 (effective Jan. 1, 2023).
[195] California Privacy Rights Act of 2020, sec. 14, § 1798.140(a)-(e).
[196] California Privacy Rights Act, sec. 4, § 1798.100(a).
[197] California Privacy Rights Act, sec. 24, § 1798.199.55(a)(2).
[198] *See Data Protection Laws of the World*, DLA PIPER, https://www.dlapiperdataprotection.com/index.html?t= world-map&c=AU (last visited Dec. 11, 2021).

demonstrate that the U.S. takes privacy seriously enough to dedicate resources to its proper adjudication. Current presiding federal agencies in the privacy space (i.e., the FTC) are simply insufficient when it comes to handling the volume of data privacy incidents in the U.S. The insufficiency is especially glaring due to the fact that the FTC does not provide a vehicle for U.S. citizens to pursue private actions when seeking recourse for breaches and violations related to the compromising of their most sensitive data.

The DPA section of the CPRA also helps to distinguish it from other laws like the previously discussed BIPA,[199] or Texas' biometric privacy law,[200] which also both explicitly regulate the storage and use of biometric data. However, despite the notably high volume of private lawsuits that have been filed in the short window between BIPA's enactment and present day,[201] along with other similar statutes, these statutes represent an overall less robust approach to data protection than the CPRA. While many other biometric privacy statutes restrict the collection and storage of biometric identifiers (e.g., iris scan, fingerprint, scan of face geometry and a voiceprint), the extent of coverage varies quite substantially.[202] Further, no other statute calls for a DPA. Thus, despite BIPA's current regulatory dominance, the CPRA is poised to overtake it due to the scope of its CPPA. The CPPA not only offers a comprehensive, well-defined set of processes and procedures for adjudication,[203] but it also creates a body that (1) is vested with full

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[199] *See* 740 ILL. COMP. STAT. ANN. 14/5 (West 2008).
[200] *See* TEX. BUS. & COM. CODE ANN. § 503.001 (West 2009).
[201] Shari C. Lewis, *Facing a Flood of BIPA Lawsuits: Lessons Can be Learned From Statute*, N.Y. L.J. (June 15, 2020, 12:32 PM), https://www.law.com/newyorklawjournal/2020/06/15/facing-a-flood-of-bipa-lawsuits-lessons-can-be-learned-from-statute/.
[202] Michael Monajemi, Privacy Regulation in the Age of Biometrics That Deal with A New World Order of Information, 25 U. MIAMI INT'L & COMPUT. L. REV. 371, 399 (2018).
[203] Benjamin B. Wagner et al., *The Evolution of Privacy Enforcement in California: CPRA and the CA Attorney General's Office*, LAW.COM: THE RECORDER (Jan. 22, 2021, 4:29 PM), https://www.law.com/therecorder/2021/01/22/the-evolution-of-privacy-enforcement-in-california-cpra-and-the-ca-attorney-generals-office/?slreturn=20210210202031.

administrative power, authority, and jurisdiction to implement and enforce the CCPA, (2) will have the power to cooperate with other privacy enforcement agencies in the state, as well as in other states, territories, and countries, (3) is expected to coordinate investigatory actions with regulators in other jurisdictions, including European data protection authorities, (4) will have an educational function and is charged with promoting public awareness and understanding of the risks, rules, responsibilities, safeguards, and rights in relation to the collection, use, sale and disclosure of personal information, and (5) will provide technical assistance and advice to the California Legislature with respect to privacy-related legislation, "monitor" the developments in the field of personal information protection, and establish a mechanism for organizations that are not subject to the CPRA to voluntarily self-certify compliance.[204]

c. Next Steps: Drafting a Federal Biometric Data Privacy Law

While the U.S. privacy regulatory environment is fragmented and dysfunctional from a compliance perspective,[205] a survey of data privacy laws globally and domestically reveals best practices that a federal law should address:

1. **Stand-Alone Policy**: Either a dedicated, standalone statute to address biometric data protection or a special carve-out in a general privacy statute that serves to distinguish the specialized sensitivity attached to biometric data privacy. All data is not created, processed, or valued equally, thus a statute that fails to capture the requisite nuance to adequately address this

[204] Lydia de la Torre & Glenn Brown, *What is the California Privacy Protection Agency?*, IAPP (Nov. 23, 2020), https://iapp.org/news/a/what-is-the-california-privacy-protection-agency/; *see generally* CAL. CIV. CODE § 1798 (West 2018).
[205] Michael Beckerman, *Americans Will Pay a Price for State Privacy Laws*, N.Y. TIMES (Oct. 14, 2019), https://www.nytimes.com/2019/10/14/opinion/state-privacy-laws.html.

will be weak and cannot be considered sufficient to be a true biometric data privacy law.

2. **Control Mechanisms**: Adequate control mechanisms that reflect the basic tenets and principles of privacy and security policy, for example, ensuring that (a) only data necessary to fulfill a specific purpose is collected (i.e., "purpose limitation"), (b) only individuals who require access to the data to fulfill that purpose are given authorization to do so, to limit access by unintended actors by reducing the attack surface, (c) sufficient security measures are adopted, both physical and onsite (e.g., keycard access to servers, robust password and employee user authentication rules, etc.) and digitally, both encryption and network monitoring, to avoid unnecessarily compromising sensitive data, and (d) any other reasonably necessary steps an organization can take to secure user information.

3. **Transparency**: A requirement for transparency via communications through (a) organizational privacy policies (both internal- and external-facing), (b) appropriate privacy notices (including data breaches) and disclosures to consumers, and (c) external guidance to prospective consumers and those potentially impacted by a product or service that discloses critical information around their data protection.

4. **Secure Infrastructure**: Secure infrastructure through sufficient staffing and/or resources (e.g., a data protection officer),[206] as well as required trainings for all employees so that they are knowledgeable in best practices.[207] Training is especially important because it serves as a way to increase the efficacy of well-crafted

[206] *See generally* Heiman, *supra* note 69; 201 MASS. CODE REGS. 17.03 (2021).
[207] *See generally* Heiman, *supra* note 69; California Privacy Rights Act of 2020, sec. 14, § 1798.140(a)-(e).

policies, as words are only as meaningful as the actions put behind them. One of the biggest risks of data breaches facing companies today is from employees who are not able to navigate common forms of attacks, like phishing. Phishing attacks are fraudulent attempts to gain access to sensitive information by impersonating a trustworthy entity (e.g., a bank or trusted person) in a digital communication (e.g., email or text).[208] Phishing is an example of a cyber-attack that is both less technically sophisticated and highly preventable with proper training.

5. **Dedicated Adjudicative Body**: A dedicated agency for providing focused adjudication. While agencies like the FTC have been instrumental in the fight to protect consumer data, not only does it fail to accommodate private lawsuits filed by individuals or groups of individuals, but it also simply cannot handle the sheer volume of data privacy incidents on a national scale the way it is currently established.[209] The courts are also insufficient for providing rulings on the volume of data privacy specific cases in terms of resources and overall capability. As such, given the sensitivity and need for a dedicated data protection agency, a fully functional biometric privacy law will likely mirror this aspect of the CPRA.

6. **Consent**: The spirit and substance of a federal biometric privacy law must have user consent embedded into its DNA, as this is arguably the most important aspect of

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[208] Josh Fruhlinger, *What Is Phishing? How This Cyber Attack Works and How to Prevent It*, CSO (Sept. 4, 2020), https://www.csoonline.com/article/2117843/what-is-phishing-how-this-cyber-attack-works-and-how-to-prevent-it.html.

[209] Chris J. Hoofnagle et al., *The FTC Can Rise to The Privacy Challenge, But Not Without Help from Congress*, BROOKINGS (Aug. 8, 2019), https://www.brookings.edu/blog/techtank/2019/08/08/the-ftc-can-rise-to-the-privacy-challenge-but-not-without-help-from-congress/.

> data protection and one of the most common provisions addressed across every current and existing privacy law. Without consent to access and process the data of those impacted by an organization's products or services, there is no trust or transparency, thus, the spectrum of efficacy of a biometric data privacy law is proportionally and intrinsically tied to its coverage of consent.

IV. CONCLUSION

2020 represented a culmination of issues that, while not necessarily novel given the U.S.' history, happened to occur at a time when society has reached a new level of conscientiousness, especially with regards to consumer data protection. The Biden administration is poised to address data privacy at a federal level and has already taken forward steps by appointing Chris Hoff, a season privacy veteran, to oversee Privacy Shield discussions to repair and reinforce U.S.-EU data privacy relations.210 Further, Senator Brian Schatz (D-Hawai'i) and a group of seventeen of his fellow senators have reintroduced a proposal for a federal privacy law focusing on, inter alia, a duty of care, loyalty, and confidentiality with regards to data.211 Even more telling has been the issuance of Biden's 2021 executive orders, first on "[i]mproving the Nation's Cybersecurity,"[212] in response to Solar Winds and other "increasingly sophisticated malicious cyber campaigns,"[213] from other bad actors in states like Russia and China. Then, a more privacy-specific executive order aimed at "[p]rotecting Americans'

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[210] Caitlin Fennessy, *Biden Appoints Christopher Hoff To Oversee Privacy Shield Talks*, IAPP (Jan. 20, 2021), https://iapp.org/news/a/biden-appoints-chris-hoff-to-oversee-privacy-shield-talks/.

[211] *See* Press Release, Brian Schatz, Senator, Hawaii, Schatz Leads Group of 17 Senators in Reintroducing Legislation to Help Protect People's Personal Data Online (Mar. 23, 2021), https://www.schatz.senate.gov/news/press-releases/schatz-leads-group-of-17-senators-in-reintroducing-legislation-to-help-protect-peoples-personal-data-online.

[212] Exec. Order No. 14028, 86 C.F.R. § 26633 (2021).

[213] *Id.*

Data from Foreign Adversaries."[214] The latter not only formally acknowledges individuals' data privacy as a "human right" but also lays out requirements for scrutinizing software that risks improper data use and collection.[215] Taken together, the Biden administration has demonstrated interest in the digital economy and is far more likely to push for federal laws encompassing the spirit of these orders. As such, it is even more important for companies and compliance officers to anticipate what will be required of them.

While the actual enactment of a federal law may be unpredictable, legislators, scholars, and corporate policymakers alike should think through the context and intent behind such a law, especially with regards to biometric data, as it is one of the most sensitive and potentially damaging classes of individuals' data. They should also seek to incorporate best practices based on existing laws and standards to ensure the right approach is being taken to achieve necessary and robust protection, and to get ahead of potentially forthcoming compliance requirement.

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[214] Exec. Order No. 14034, 86 C.F.R. § 31423 (2021).
[215] *Id.*

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